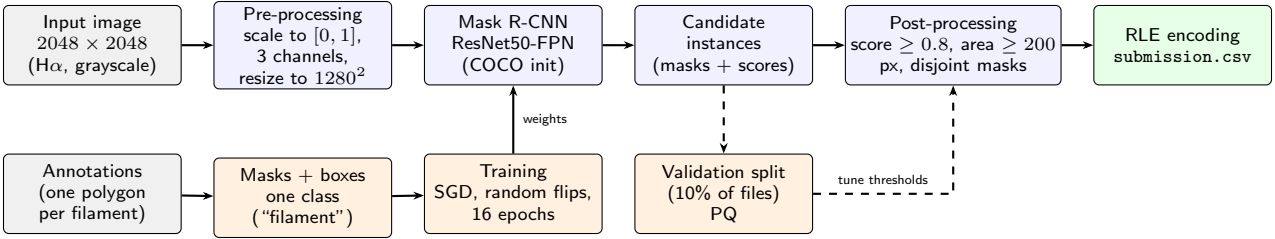
Input image
$2048 \times 2048$
(H$\alpha$, grayscale)
Pre-processing
scale to $[0, 1]$,
3 channels,
resize to $1280^2$
Mask R-CNN
ResNet50-FPN
(COCO init)
Candidate
instances
(masks + scores)
Post-processing
score $\geq 0.8$, area $\geq 200$
px, disjoint masks
RLE encoding
submission.csv
weights
Annotations
(one polygon
per filament)
Masks + boxes
one class
("filament")
Training
SGD, random flips,
16 epochs
Validation split
(10% of files)
PQ
tune thresholds